The Feldstein–Horioka Puzzle Is Back

Saving and investment came apart before 2008. They have moved back together since

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Abstract

In 1980 Feldstein and Horioka found that countries that save more also invest more, almost one for one, as if capital could not cross borders. By the mid-2000s the link had nearly vanished, and in the euro area it was declared dead. We update the classic regression with World Bank data for 1980–2024. The link fell to near zero in 2000–07 and has since returned: across OECD countries, an extra point of saving now goes with about half a point of extra investment at home, the same as in the 1980s. The rebound holds in every sample and measure we try but one, a worldwide sample using domestic saving, where it is weaker. It reflects much smaller current-account imbalances, not a change in saving.

1 The question and the regression

If capital moved freely, a country's investment would not depend on its own saving: a high-saving country would lend abroad and a low-saving one would borrow. Feldstein and Horioka (1980) regressed the investment rate on the saving rate across 16 OECD countries, using averages over 1960–74, and found a slope close to 0.9. Their reading was that most saving stays at home. The "puzzle" became one of the six major puzzles in international macroeconomics (Obstfeld and Rogoff, 2000).

We run the same cross-country regression for successive periods: the period-average ratio of gross capital formation to GDP on the period-average ratio of gross national saving to GDP, one observation per country, with heteroskedasticity-robust standard errors. Data are the World Development Indicators (update of October 2026). A country enters a period if it has at least 70% of the years. National saving is the relevant concept because the current account equals national saving minus investment. The slope β measures how much of a country's saving above the cross-country average shows up as investment above the average. Since the current account is $S - I$, $\beta = 1 - \mathrm{cov}(CA, S)/\mathrm{var}(S)$: the slope falls when high savers run large surpluses.

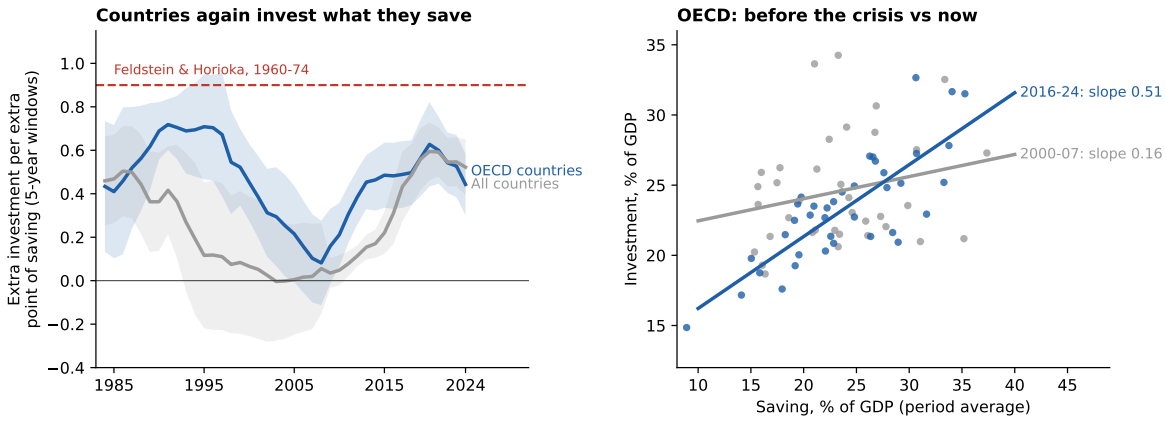


Figure 1: Left: slope of investment on saving across countries, estimated on rolling five-year averages ending in the year shown, with 95% bands. OECD: the 38 current members. All countries: population above one million. Right: OECD period averages of saving and investment (% of GDP), with fitted lines.

2 The answer

The slope fell from 0.56 in the 1980s and 0.65 in the 1990s to 0.16 (s.e. 0.11) in 2000–07 across the OECD, and to zero across all countries. It has since come back to 0.51 (s.e. 0.06) in the OECD and 0.54 (0.06) worldwide for 2016–24 (Figure 1). In the euro area, where Blanchard and Giavazzi (2002) saw "the end of

	1980–89	1990–99	2000–07	2016–24
OECD, 38 countries	0.56 (0.14)	0.65 (0.10)	0.16 (0.11)	0.51 (0.06)
OECD, balanced panel of 23	0.56 (0.15)	0.67 (0.11)	0.19 (0.13)	0.47 (0.08)
OECD, excl. financial centers and oil exporters	0.46 (0.24)	0.73 (0.09)	0.29 (0.12)	0.50 (0.08)
same, domestic instead of national saving	0.42 (0.18)	0.51 (0.15)	0.30 (0.13)	0.53 (0.09)
same, fixed investment only	0.42 (0.20)	0.69 (0.08)	0.26 (0.11)	0.48 (0.08)
Euro area, excl. Ireland and Luxembourg	0.69 (0.18)	−0.26 (0.42)	−0.04 (0.17)	0.41 (0.12)
All countries, population > 1 million	0.43 (0.12)	0.13 (0.18)	0.00 (0.13)	0.54 (0.06)
excl. financial centers and oil exporters	0.53 (0.12)	0.43 (0.11)	0.32 (0.08)	0.60 (0.06)

Table 1: Slope of period-average investment on period-average saving (% of GDP), robust standard errors in parentheses. Financial centers: Ireland, Luxembourg, Switzerland, Singapore, Hong Kong, Malta, Cyprus. Oil exporters: Norway and 21 others. The 2008–15 slopes lie in between (0.43 for the OECD).

the Feldstein–Horioka puzzle", the slope went from slightly negative in 2000–07 to 0.55 in 2016–24 for all members, and to 0.41 excluding Ireland and Luxembourg (Table 1).

Dropping financial centers and oil exporters, holding the country set fixed, measuring saving before factor income and transfers, or counting only fixed investment changes the level of the pre-crisis trough but not the rebound. The one exception is the worldwide sample with domestic saving (0.05 in 2000–07, 0.19 in 2016–24), where remittance-receiving economies make domestic and national saving very different.

The mechanism is visible in the current accounts. Across OECD countries, the standard deviation of the period-average current account rose from 2.5% of GDP in the 1980s to 5.9% in 2000–07 and has fallen back to 3.9%. The spread of saving rates barely moved (5.2% to 6.1%). High savers no longer send as much of their saving abroad.

3 What this does and does not show

The slope is a summary of how far countries' investment tracks their saving. It is not a clean measure of capital mobility. A country can keep its current account small by choice, by policy, or because lenders are cautious after a crisis, even when capital is free to move. The data cannot tell these apart. What they show is that the pre-crisis world, in which saving and investment had nearly decoupled, did not last.

4 Why it matters

When a government or a country saves less, the question is whether its investment falls with it or foreign lenders fill the gap. Before 2008 the answer looked like "foreigners fill it". Since then, across countries, about half of a gap in national saving shows up as a gap in investment at home. For deficits, pension reform and industrial policy alike, national saving matters again.

Notes. World Bank series NY.GNS.ICTR.ZS, NE.GDI.TOTL.ZS, NY.GDS.TOTL.ZS, NE.GDI.FTOT.ZS and BN.CAB.XOKA.GD.ZS. Saving data are thin before 1980, so we cannot reproduce the original 1960–74 sample; the 0.9 benchmark is Feldstein and Horioka's own estimate. An IMF cross-check was not possible. We have not checked for other recent updates beyond a short search.

References. Blanchard, O. and F. Giavazzi (2002), Current account deficits in the euro area: the end of the Feldstein–Horioka puzzle?, *Brookings Papers on Economic Activity* 2002(2), 147–209. Feldstein, M. and C. Horioka (1980), Domestic saving and international capital flows, *Economic Journal* 90(358), 314–329. Obstfeld, M. and K. Rogoff (2000), The six major puzzles in international macroeconomics: is there a common cause?, *NBER Macroeconomics Annual* 15, 339–390.